AI-REVIEWED WORKING BRIEFING · TECHNICAL REFERENCE V1.3

EnergyWave Physical Architecture & Baseline Model

Joint Engineering Baseline & Experimental Protocol (Antigravity, GPT/Codex, Claude, Ian McLean)

DATE	ORIGINATORS	RECIPIENT	STATUS
October 5, 2026	Antigravity & GPT/Codex	Claude Desktop & Ian McLean	AI-Reviewed Working Briefing v1.3

> **Working Status & Scope:** This document is an AI-reviewed working briefing establishing physical accounting, metrology constraints, and an unconditioned baseline experiment. It does not establish scientific consensus or experimental validation. Its goal is to replace unverified assumptions with rigorous, testable physics and provide an empirical path to evaluate Ian McLean's specimen once the circuit and conditioning process are defined.

1. Foundational Ground Rules & First-Principles Accounting

- **First Law Invariance:** Energy cannot be created from nothing. The device must satisfy closed thermodynamic accounting: `dU/dt = P_all_inputs − P_load − P_loss`.
- **Distinguish Proven Conversion from Unverified Sources:** The documented electrical conversion chain (resonant LC tank, rectifier/charge pump, DC-link buffer, adaptive feedback controller) is standard conversion engineering. The physical mechanism that supplies energy above the conventional baseline remains an unestablished hypothesis.
- **Measurement Error vs. Physical Source:** Instrumentation errors (e.g., probe timing deskew) explain an *apparent* power surplus on an oscilloscope; they do not physically supply energy.
- **Ordinary Power Inputs:** In addition to novel hypotheses, ordinary excitation power, bench supplies, controller power, and internal capacitive/battery storage must be completely metered.

2. Complete Energy Accounting & Efficiency Formulation

Governing Energy Conservation Law:

$$E_{\text{all inputs}} = E_{\text{load}} + E_{\text{heat and other outputs}} + \Delta U_{\text{device}}$$

Here, U includes energy stored in capacitors, batteries, magnetic states, and internal thermal reservoirs. For steady operation where internal stored energy remains unchanged ($\Delta U_{\text{device}} = 0$), the appropriate conversion efficiency is:

$$\eta = P_{\text{load}} / P_{\text{total external input}}$$

Accounting Note: Measured total external input already covers internal losses ($P_{in} = P_{load} + P_{loss}$). Adding internal losses to the denominator would count them twice.

The Closed-Loop Feedback Reality

If the intended mechanism is feeding the rectified output back into its own excitation: with forward conversion efficiency $\eta_{fwd} < 1$ (e.g., 0.90) and feedback fraction $\beta \le 1$, the loop gain is $T = \beta \cdot \eta_{fwd} < 1$. The recirculating power decays as a geometric series:

$$P_n = P_0 \cdot (T)^n \to 0 \quad (\text{as } n \to \infty)$$

Self-sustained operation without an independent input or depletion of stored internal energy cannot occur under classical or modern electrodynamics.

3. What Could "Conditioning" Physically Do? Hypotheses to Test

In historical literature referenced by the project (e.g., Floyd Sweet's Vacuum Triode Amplifier, cited on page 5 of `EnergyWave_Deck.pdf`), barium or strontium ferrite permanent magnets undergo an electrical "conditioning" process. The physical nature of this change remains unestablished. We formulate the hypotheses to test:

Hypothesis 1: Finite Stored Energy Depletion (Hypothetical Magnetic/Strain State Release)

Physical Hypothesis: Conditioning might induce a non-equilibrium domain-wall configuration, exchange bias, or internal mechanical strain. External excitation could trigger domain relaxation, releasing finite stored energy.

Hysteresis & Spatial Integration: For a hysteretic ferromagnetic material, magnetic work involves path-dependent hysteresis and irreversible dissipation; it cannot automatically be equated to recoverable stored energy or electrical output. The cyclic energy dissipated as heat requires spatial integration over the core volume (see MIT 6.061):

$$E_{cycle} = \int_{V_{core}} \oint \mathbf{H}(\mathbf{r}) \cdot d\mathbf{B}(\mathbf{r})\, dV \qquad [\text{Joules}]$$

Where local fields **H**(**r**) and **B**(**r**) vary spatially across the 3D core geometry. Multiplying a single scalar B-H loop by total volume assumes uniform field distribution throughout the material.

Distinguishing $(BH)_{max}$ from Stored Energy: The maximum energy product $(BH)_{max}$ is a static figure of merit for permanent magnet design (characterizing the maximum magnetic energy density available in an external air gap; see MIT 6.685). $(BH)_{max}$ cannot be multiplied by core volume to define a total recoverable thermodynamic "battery" capacity or predict operating discharge duration.

Material-State Indicators: Remanence (B_r) and coercivity (H_c) are material-state indicators. Changes in B_r or H_c can arise from multiple mechanisms (such as AC demagnetization or thermal relaxation). Observing lower remanence does not alone prove the core powered the circuit; attribution requires quantitatively connecting measured material-state changes to the complete energy balance.

Hypothesis 2: Open-System Environmental Transduction (Ambient Energy Harvesting)

Physical Hypothesis: The device acts as an open thermodynamic system, capturing energy from the surrounding environment.

- **Thermal Fluctuations (Brillouin's Paradox & The Second Law):** At a single equilibrium temperature, a passive rectifier *cannot* continuously extract useful electrical work from Johnson-Nyquist thermal noise. This restriction follows from including the rectifier's own thermal fluctuations (as shown by Léon Brillouin) rather than treating it as a noiseless ideal diode. To deliver sustained thermal work, there must be a maintained temperature difference ($T_{hot} - T_{cold} > 0$) or another non-equilibrium driving source, strictly bounded by Carnot efficiency $\eta \leq 1 - T_{cold}/T_{hot}$. Temperature difference alone does not measure heat flow or complete the energy balance.
- **Ambient Electromagnetic Flux:** Ambient RF harvesting is a separate physical mechanism governed by incident Poynting flux $\iint (\mathbf{E} \times \mathbf{H}) \cdot d\mathbf{A}$. Evaluating ambient RF requires measured field strengths, frequencies, and antenna/core coupling. An illustrative 1 $\mu W/cm^2$ flux incident across a 150 cm^2 geometric area represents 150 µW of incident power; actual absorbed power requires characterized electromagnetic coupling and impedance matching.

Decisive Testing Requirements: Testing environmental coupling requires an RF-shielded enclosure with characterized attenuation (SE in dB across frequency) and a calibrated calorimeter to measure actual heat flow rates.

Hypothesis 3: High-Q Metrology Phase Skew (Apparent Power vs. Real Power)

Physical Mechanism: Timing deskew errors between voltage and current probes in high-Q or highly reactive circuits alter the measured phase angle θ, severely distorting the computed real power (see Tektronix Application Note). Measurement error explains an *apparent* surplus; it does not physically supply energy.

$$P_{measured} = V_{RMS} \cdot I_{RMS} \cdot \cos(\theta + 2\pi f \Delta t)$$

Explicitly Hypothetical Sensitivity Benchmark (GPT/Codex Benchmark):
Consider a hypothetical circuit carrying 10 VA of apparent power with a true real power dissipation of **+50 mW** (true phase θ = arccos(0.050 / 10) = 89.7135° = 1.565795 rad). With an adverse timing deskew error of **Δt = 2 nanoseconds**:

The calculated power reading crosses zero at approximately:

$$f_{zero} = (\pi/2 - \theta) / (2\pi \Delta t) \approx 0.005001 / [2\pi (2 \times 10^{-9})] \approx \mathbf{397{,}889\ Hz\ (\approx 398\ kHz)}$$

Frequency (f)	Phase Error (2π f Δt)	Measured Power Reading	Metrological Impact
1 kHz	+0.00072°	**+49.87 mW**	Negligible (< 0.3% error)
100 kHz	+0.0720°	**+37.43 mW**	25.1% under-reporting of consumed power
398 kHz	+0.2865°	**0.00 mW**	**Calculated zero-crossing threshold**
1 MHz	+0.7200°	**-75.66 mW**	**Sign reversal! False "generation" reading!**

Critical Metrology Takeaway: Sign reversal is possible, but depends strictly on frequency, phase, and signal magnitude (crossing zero at ≈ 398 kHz in this benchmark; 1 MHz is a sampled point above reversal, not a unique threshold). Calibration must cover the complete measurement chain; aligning probe zero-crossings alone is insufficient.

4. The Proposed Baseline Experiment: Four-Step Staged Protocol

The AI Council jointly agrees to **begin with an unconditioned characterization baseline**. Rather than testing a complex circuit with feedback and rectification, we start with an externally driven core-and-coil assembly with a fixed resistive load on the pickup winding:

1. **Step 1 — Identify One Physical Specimen:** Select one concrete hardware specimen: core material, exact geometry, magnetic orientation, and initial state.
2. **Step 2 — Characterize Before Conditioning:** Measure winding DC/AC resistance, impedance, coil coupling, and temperature-dependent magnetic response. *Crucial rule:* Because characterization drives may themselves alter the core's magnetic state, drive levels must be recorded and initial/final states monitored.
3. **Step 3 — Externally Driven Loaded Measurements:** Select excitation parameters from the specimen's characterized operating range. Deliver power to a fixed non-inductive resistive load. Simultaneously integrate input and output power:

$$E_{exc} = \int v_{exc}(t)\, i_{exc}(t)\, dt \qquad E_{load} = \int i_{load}^2(t)\, R_{load}\, dt$$

4. **Step 4 — Conditioning as a Separate Experiment:** Introduce conditioning only after the unconditioned baseline is fully mapped. The conditioning experiment must define its intended physical state change, meter total conditioning energy ($E_{cond} = \int v_{cond}\, i_{cond}\, dt$), and compare results against matched control specimens receiving a sham treatment.

> **The First Testable Question:**
> *"Does a defined conditioning procedure produce a repeatable change in magnetic response or loaded conversion performance compared with an appropriate control?"*
> An improvement in coupling or reduction in core loss is valuable engineering even if all output comes from excitation. A claim of additional energy requires full closure: $E_{all\ inputs} = E_{load} + E_{other\ outputs} + \Delta U_{device}$.

5. Candidate Hardware Parameters for Proposed Baseline

Note: The parameters below represent proposed experimental choices to enable preliminary loss estimates. They are unvalidated choices pending Ian McLean's confirmed specimen design:

Subsystem	Proposed Experimental Choice	Technical Role & Validation Requirement
Core Material	Sintered Strontium Ferrite ($SrFe_{12}O_{19}$) Grade 8 Permanent Magnet	Tests Sweet VTA permanent magnet lineage. Control specimens must be matched in material, geometry, and preparation. (Soft ferrite serves as an instrumentation reference, not a material control).
Core Dimensions	Toroid (OD 75 mm, ID 50 mm, H 20 mm) or Block (150 × 100 × 25 mm)	Defines magnetic path length ℓ_e and cross-sectional area A_e.
Excitation (N_1)	50 turns, AWG 20 enameled copper wire	Proposed choice. Winding resistance, AC losses, and thermal limits must be validated against measured current and duty cycle.
Pickup (N_2)	200 turns, AWG 24 enameled copper wire	Proposed choice. Step-up ratio provides measurable loaded voltage across test resistor.
Test Load (R_{load})	50 Ω non-inductive power resistor (Kelvin-sensed)	Provides unambiguous purely resistive load ($\cos\phi = 1.00$). Treat as component with measured impedance & temperature dependence.

Subsystem	Proposed Experimental Choice	Technical Role & Validation Requirement
Drive Source	Linear RF Power Amplifier driven by synthesized generator	Low THD (< 0.1%) sine excitation, 10 kHz – 100 kHz.
Conditioning	**Conditioning remains unspecified.**	**No capacitor-discharge, saturation-soak or mechanical protocol has been selected. First establish the specimen's conventional response. Any subsequent conditioning experiment must define its intended state change and account for conditioning, operation and restoration energy.**

6. Counsel, Docket, and Corporate Formation Status

Workstream	Verified Status	Action Required / Next Step
K&L Gates Patent Inquiry	• Dennis Majewski confirmed K&L Gates has 0 filings. • Verified historical filing exists: **U.S. Provisional Application No. 63/763,998** (Filed Feb 27, 2025; Missing-parts notice Mar 5, 2025; Electronic payment receipt May 8, 2025). • May 5 correction document has blank signature line (submission/acceptance unconfirmed).	Draft reply in Outlook requests counsel to pull Patent Center docket, confirm statutory bars, and provide intake checklist/fee estimate.
Corporate Entity Formation	• Corporate counsel Trinh & Jason (K&L Gates) requested Delaware formation details on Sept 28. • Assignment of domain (`energywave.ai`) and IP is strictly gated on Delaware formation.	Send formation packet to Trinh & Jason to establish EnergyWave, Inc. so assignment agreements can execute.
Aedan Looking Glass (ALG) Agreement	• Sept 16 ALG agreement shows Gary's signature, but Ian's contractor line is blank. • Integrated Protective Rider has Ian's signature, but company line is blank. • Exhibit A lists conflicting provisional number `63/763,986` (scrivener's error vs. receipt `...`998).	Counsel review required to formally settle IP carve-outs and reconcile patent numbering.

7. Primary Technical References

- **MIT 6.685:** Permanent-magnet energy product and magnetic-circuit interpretation: Chapter 6 (PDF)
- **MIT 6.061:** Magnetic work, hysteresis, and loss accounting: Chapter 8 (PDF)
- **Tektronix Application Note:** Voltage/current probe deskew and power measurement in power supplies: Power Supply Measurement & Analysis

8. Immediate Concrete Decisions for Ian McLean

1. **Identify Candidate Physical Specimen:** Confirm whether to baseline on a specific off-the-shelf Grade 8 Strontium Ferrite magnet block/toroid or specify Ian's actual laboratory core.
2. **Confirm Proposed Unconditioned Baseline Protocol:** Approve Step 1–3 of the proposed experiment (unconditioned pre-characterization + externally driven loaded test) prior to introducing any conditioning tests.
3. **Corporate Formation Authorization:** Confirm release of Delaware formation responses to Trinh and Jason to unblock corporate assignment.